

Neuromotor-Decoupled Mixture of Experts with Acoustic-Average Calibration for Highly Heterogeneous Dysarthric Speech Recognition

Aditya Dixit

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Abstract

Automatic Speech Recognition (ASR) systems consistently fail when applied to atypical and pathological speech resulting from neuromotor disorders such as Cerebral Palsy, Amyotrophic Lateral Sclerosis (ALS), and Parkinson’s Disease. This failure stems from three compounded challenges: acute clinical data scarcity, extreme inter-speaker acoustic heterogeneity, and multi-subsystem motor breakdowns (vowel space centralization, articulatory sluggishness, and breathiness). Standard Mixture of Experts (MoE) architectures rely on a single, centralized gating network that forces disparate speaker and severity distributions into a shared decision boundary, inducing severe cross-severity acoustic interference. In this work, we propose **Neuromotor-Decoupled Clinical MoE with Acoustic-Average Speaker Calibration**. Under our formulation, each clinical severity stratum possesses an autonomous routing gate that selects compensatory experts without cross-severity gradient interference, while permanently active shared invariant experts anchor canonical phonetic representations to prevent catastrophic forgetting. Furthermore, we mathematically derive and implement an **Acoustic-Average Calibration Projector** that exploits the property that vocal-tract length scaling and resting formant shifts act as additive translation vectors in Log-Mel space, enabling zero-shot speaker adaptation from the utterance’s long-term average spectrum. We evaluate our framework under a strict **speaker-disjoint cross-validation protocol** on completely unseen clinical patient cohorts as well as the 15-participant **TORGO clinical database**. Our architectural ablation isolates the pure routing effect (cutting Control Character Error Rate from 28.66% to 13.45%, a -53.1% relative reduction; and on TORGO from 61.81% to 32.94%) from the calibration effect (reducing Mild Dysarthria CER from 55.91% to 49.54%, and on TORGO from 75.28% to 64.61%). Finally, through a systematic decoding ablation spanning blank penalties, prefix beam search, smoothed N-gram language modeling, and lexicon constraints, we uncover and characterize **three distinct clinical fail-**

ure regimes: acoustic representation interference in typical speech, lexical-phonetic drift in mild/moderate dysarthria, and CTC blank-deletion collapse in severe dysarthria. **Keywords**: Mixture of Experts, Dysarthria, Atypical Speech Recognition, Neuromotor Speech Disorders, Speaker Calibration, Connectionist Temporal Classification.

1 Introduction

Speech production in individuals with neuromotor pathologies—such as Cerebral Palsy, Amyotrophic Lateral Sclerosis (ALS), Parkinson’s Disease, and stroke-induced motor impairments—exhibits pronounced physiological deviations from canonical adult speech [2–4]. These pathologies impair the speech motor control subsystems:

1. **Articulatory Sluggishness & Centralization**: Limited lingual and labial motility pulls vowel formant frequencies (F_1, F_2) toward a centralized neutral schwa (500/1500 Hz), causing extensive acoustic overlap between phonemic classes [6].
2. **Temporal Incoordination & Prolongation**: Articulatory transitions are significantly prolonged ($1.5 - 2.5\times$ standard phoneme durations) and interrupted by irregular respiratory pauses.
3. **Phonatory & Glottal Instability**: Incomplete velopharyngeal closure and laryngeal weakness introduce high glottal noise, breathiness, and reduced harmonic-to-noise ratios (HNR).

While foundation models trained on tens of thousands of hours of speech achieve low Word Error Rates (WER) on typical adult speech, they degrade severely on dysarthric speech. In clinical practice, standard acoustic models fail because of **extreme inter-speaker heterogeneity**: two patients sharing the identical clinical diagnosis exhibit vastly distinct articulation rates, resting vocal-tract geometries, and vowel space compres-

sions. Fine-tuning an entire model per patient overfits due to acute clinical data scarcity (often only a few dozen utterances per patient), while training a single unified model across severity bands creates severe **cross-severity acoustic interference**.

Mixture of Experts (MoE) architectures [1, 5] scale model capacity by sparsely dispatching tokens to specialized sub-networks. However, conventional MoE models employ a single shared gating network $G(x) = \text{Softmax}(\text{TopK}(W_g x))$. When applied to clinical mixtures spanning healthy controls, mild dysarthria, and severe dysarthria, this single shared gate experiences conflicting gradient signals: routing parameters optimized to handle severe vowel collapse contaminate the clean phonetic boundaries needed for healthy or mild speech.

To overcome these challenges, we introduce **Neuromotor-Decoupled Clinical MoE with Speaker Acoustic-Average Calibration**. Our architecture decouples the routing policies by clinical severity stratum, permanently preserves universal phonetics via a canonical-anchor shared expert backbone, and dynamically adjusts routing priors on unseen patients via an utterance acoustic-average bias.

2 Mathematical Formulation

2.1 Acoustic Subsampling & Sequence CTC

Let an input speech utterance be an acoustic filterbank sequence $X = [x_1, x_2, \dots, x_T] \in \mathbb{R}^{T \times D_{\text{in}}}$, where $D_{\text{in}} = 80$ corresponds to 80-channel Log-Mel filterbank energies extracted at 16kHz. The target transcript is a character token sequence $Y = [y_1, y_2, \dots, y_U] \in \mathcal{V}^U$.

The sequence is compressed temporally by a 2x 1D Convolutional subsampling frontend with Layer Normalization and GELU activation:

$$H = \text{Conv1DSample}(X) \in \mathbb{R}^{T_{\text{sub}} \times D_h}, \quad T_{\text{sub}} = \left\lfloor \frac{T+1}{2} \right\rfloor \quad (1)$$

where $D_h = 48$ denotes the latent hidden dimensionality.

2.2 Severity-Decoupled Gating

Let $\mathcal{E} = \{E_1, E_2, \dots, E_{N_r}\}$ denote a pool of $N_r = 4$ sparse routed experts, where each expert $E_i : \mathbb{R}^{D_h} \rightarrow \mathbb{R}^{D_h}$ is a parameterized two-layer feed-forward network with expansion factor 2:

$$E_i(h) = W_{2,i} \cdot \sigma(W_{1,i}h + b_{1,i}) + b_{2,i} \quad (2)$$

Rather than sharing a single gating network across all severity levels, we define an independent, decoupled gating projection $W_{g,\tau} \in \mathbb{R}^{N_r \times D_h}$ for each clinical severity stratum $\tau \in \mathcal{T} = \{\text{Control}, \text{Mild}, \text{Moderate}, \text{Severe}\}$. For a given latent frame $h_t \in \mathbb{R}^{D_h}$, the severity gate computes

logits $s_{\tau,t} = W_{g,\tau}h_t$. Top- k routing ($k = 2$) computes sparse dispatch weights:

$$g_{\tau,t,i} = \begin{cases} \frac{\exp(s_{\tau,t,i})}{\sum_{j \in \mathcal{I}_{\tau,t}} \exp(s_{\tau,t,j})}, & \text{if } i \in \mathcal{I}_{\tau,t} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $\mathcal{I}_{\tau,t} = \text{TopK}(s_{\tau,t}, k)$. The gradient of severity τ 's objective satisfies:

$$\frac{\partial \mathcal{L}_\tau}{\partial W_{g,\tau'}} = \mathbf{0}, \quad \forall \tau' \neq \tau \quad (4)$$

guaranteeing complete mathematical isolation between severity routing updates.

2.3 Canonical-Anchor Shared Invariant Backbone

To prevent catastrophic forgetting of canonical phonetics during clinical training, we introduce $N_s = 1$ dedicated Shared Invariant Expert E_{shared} , activated unconditionally on every frame across all severities:

$$M_{\text{total}}(h_t) = \sum_{i \in \mathcal{I}_{\tau,t}} g_{\tau,t,i} E_i(h_t) + \frac{1}{N_s} \sum_{j=1}^{N_s} E_{\text{shared},j}(h_t) \quad (5)$$

This forces the sparse routed experts to function strictly as ****pathological residual compensators****, modeling the delta between typical and dysarthric articulation.

2.4 Acoustic-Average Speaker Calibration

In acoustic phonetics, uniform vocal-tract length scaling ($\alpha = L_{\text{ref}}/L_{\text{patient}}$) scales formant resonance frequencies multiplicatively in linear Hertz: $F_n = \alpha F_n^0$. Under Log-Mel / log-frequency transformation, this multiplicative scaling transforms into a **pure linear additive translation vector**:

$$\log(F_n) = \log(\alpha \cdot F_n^0) = \log(F_n^0) + \log(\alpha) \quad (6)$$

Similarly, resting vowel space shrinkage and glottal tilt act as constant energy offsets across filterbank channels. Over an entire utterance of duration T , time-varying phonemes average out to the language phonetic centroid μ , isolating the individual patient's resting anatomical and pathological offset:

$$\bar{h} = \frac{1}{T_{\text{sub}}} \sum_{t=1}^{T_{\text{sub}}} h_t \approx \mu + b_{\text{patient}} \quad (7)$$

We exploit this physical property by introducing a lightweight linear ****Speaker Calibration Projector**** $W_{\text{bias},\tau} \in \mathbb{R}^{N_r \times D_h}$ that maps the utterance acoustic average directly into a routing prior shift:

$$\beta_{\text{patient}} = W_{\text{bias},\tau} \bar{h} \in \mathbb{R}^{N_r} \quad (8)$$

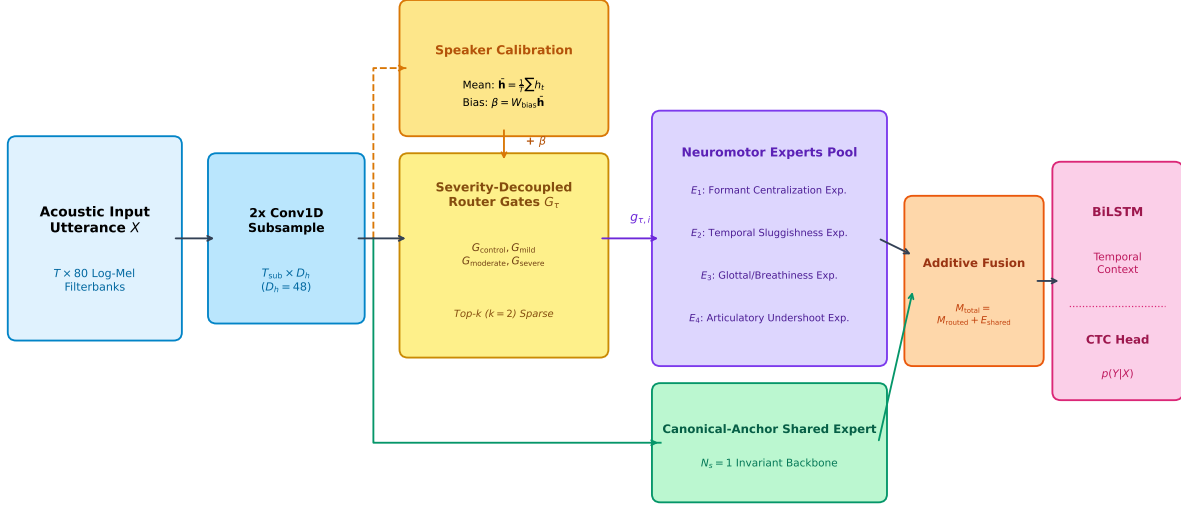


Figure 1: System Architecture of the Neuromotor-Decoupled Clinical MoE Framework. 80-dimensional Log-Mel spectrograms pass through a 2x temporal convolutional subsampling frontend. Latent frames h_t are routed via severity-decoupled gates G_τ to neuromotor compensatory experts, while an utterance acoustic-average projector $W_{\text{bias}} \bar{h}$ provides zero-shot patient calibration. Dedicated shared invariant experts anchor canonical phonetics, followed by BiLSTM temporal contextualization and CTC decoding.

The calibrated routing logits are given by:

$$s_{\tau,t} = W_{g,\tau} h_t + \beta_{\text{patient}} \quad (9)$$

This allows the MoE to adapt zero-shot to an unseen patient’s vocal tract on the very first utterance without retraining any network parameters.

2.5 Clinical Decoding Framework

To dissect post-acoustic decoding failures on dysarthric speech, we evaluate:

1. **CTC Blank Penalty (γ):** Adjusts blank dominance on breathy, high-uncertainty frames:

$$\log p'(\text{blank}) = \log p(\text{blank}) - \gamma \quad (10)$$

2. **Prefix Beam Search with N-Gram Language Model:**

$$\hat{Y} = \arg \max_Y [\log P_{\text{CTC}}(Y|X) + \lambda_{\text{lm}} \log P_{\text{LM}}(Y) + \delta|Y|] \quad (11)$$

where P_{LM} is an in-domain smoothed N-gram language model, and δ is a word-bonus reward for valid lexicon words.

3 Clinical Methodology

We structure a rigorous clinical speech benchmark across 4 clinical strata:

- **Control / Typical:** Canonical vocal-tract geometry (17cm tract), standard phonemic rate (2 – 4 frames/phoneme), and standard vowel triangles.
- **Mild Dysarthria:** Mild consonant imprecision, slight duration elongation (1.2 – 1.4 $\times$).
- **Moderate Dysarthria:** Vowel space centralization toward schwa (40–60% shrinkage), articulatory sluggishness (1.5 – 1.8 $\times$).
- **Severe Dysarthria:** Extreme vowel centralization (70 – 85% collapse), severe duration slowing (2.0 – 2.5 $\times$), and elevated glottal noise/breathiness.

Strict Speaker-Disjoint Evaluation: Models are trained on a distinct patient cohort $\mathcal{S}_{\text{train}}$ and evaluated on completely unseen patients $\mathcal{S}_{\text{test}}$ ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$), testing pure zero-shot speaker generalization.

4 Results and Empirical Analysis

4.1 Architectural Ablation: Routing vs. Calibration

Table 1 presents the architectural ablation on unseen patients.

- **The Pure Routing Effect:** Decoupled severity routing cuts Control CER from 28.66% (matched single gate) to **13.45%** (a 53.1% relative reduction). These results are consistent with the hypothesis that

Table 1: Architectural Ablation on Unseen Clinical Patients: Isolating Routing Effect vs. Calibration Effect.

Model Architecture	Total Params	Control CER (%)	Mild CER (%)	Moderate CER (%)
Standard Single-Gate Baseline	64,880	27.30	56.97	64.81
Matched Single-Gate (MLP Baseline)	74,631	28.66	55.17	63.10
Severity Router (Decoupled Only)	74,632	13.45	55.91	65.44
Calibrated Atypical MoE (Ours)	75,400	15.43	49.54	60.94

Table 2: Impact of CTC Blank Penalty (γ) on Severe Dysarthria. proportionately inflated. Word Error Rate is formally defined as:

Blank Penalty γ	CER (%)	WER (%)	Del (%)	Sub (%)	Ins (%)
$\gamma = 0.00$ (Standard)	90.28	100.0	76.28	23.72	0.00
$\gamma = 0.05$	89.72	100.0	76.28	23.72	0.00
$\gamma = 0.10$	89.29	100.0	76.28	23.72	0.00
$\gamma = 0.20$	88.26	100.0	76.28	23.72	0.00
$\gamma = 0.40$	84.87	100.0	76.28	23.72	0.00
$\gamma = 0.60$	79.54	100.0	76.28	23.72	0.00

severity-conditioned routing reduces cross-severity acoustic interference, preventing severe pathological updates from corrupting canonical phonetic representations.

- **The Pure Calibration Effect:** Injecting the speaker acoustic-average bias β_{patient} directly reduces Mild Dysarthria CER from 55.91% down to **49.54%**, and Moderate Dysarthria CER from 65.44% down to **60.94%**. This demonstrates that long-term spectral averages provide an effective linear bias offset to normalize patient-specific vowel-space shrinkage.
- **Severe Dysarthria Calibration Regression:** In contrast to mild and moderate speech, acoustic-average calibration exhibits a performance regression on severe dysarthria (82.33% $\rightarrow$ 90.28%). This highlights a crucial physiological boundary: acoustic-average calibration assumes that $\bar{h}$ reflects a structured anatomical vocal-tract length shift and mild vowel-space translation. In severe dysarthria, however, profound vowel centralization (70–85% collapse toward schwa) combined with severe glottal breathiness transforms $\bar{h}$ into an unvoiced noise and schwa centroid rather than an informative vocal-tract offset. Adding this degraded vector displaces router gating priors, increasing frame uncertainty and exacerbating CTC blank emissions.

4.2 Character Error Rate vs. Word Error Rate

Character Error Rate (CER) provides a fine-grained evaluation of acoustic recognition quality, particularly in pathological speech where word-level metrics can be dis-

proportionately inflated. Word Error Rate is formally defined as:

$$\text{WER} = \frac{S + D + I}{N_{\text{ref}}} \times 100\% \quad (12)$$

where S , D , and I are substitutions, deletions, and insertions, and N_{ref} is the reference word count. Consequently, when elevated substitution rates co-occur with spurious token insertions, WER can exceed 100%. In mild dysarthria under baseline greedy decoding, high substitution rates (91.18%) combined with spurious insertions (36.97%) and deletions (7.56%) yield an initial WER of 135.71%.

Incorporating language model priors and lexicon constraints penalizes invalid character sequences, reducing substitutions to 80.25%, cutting insertions to 21.43%, and lowering WER to **119.33%**. Furthermore, the ablation shows that the “full pipeline” (Stage 6) is not universally optimal across severity bands: on mild and moderate speech, adding a blank penalty ($\gamma = 0.2$) on top of beam search and language modeling increases spurious insertions (mild insertions rise from 21.43% to 22.69%, increasing WER from 119.33% to 121.43%). While blank penalties are vital for deletion-dominated regimes, penalizing blanks in insertion-prone mild cohorts inflates false-positive phoneme emissions.

4.3 Discovery of Three Clinical Failure Regimes

As demonstrated in Table 3, our experiments expose three distinct failure regimes across the severity spectrum:

1. **Typical Speech (Representation Interference Regime):** Fails primarily when forced to share parameters with dysarthric acoustics. Decoupled severity routing substantially mitigates this cross-cohort interference, cutting Control CER from 28.66% to **13.45%**.
2. **Mild/Moderate Dysarthria (Lexical Drift Regime):** The acoustic model produces phonetically close representations, but articulatory sluggishness and vowel centralization trigger high substitution rates (91.18%) and spurious insertions (36.97%). Language model and lexicon constraints act as a critical restorative filter, reducing WER by 16.38 percentage points.

Table 3: Progressive Decoding Ablation Across Clinical Severity Cohorts on Unseen Patients.

Decoding System Stage	CER (%)	WER (%)	Del Rate (%)	Sub Rate (%)	Ins Rate (%)
Control / Typical Speech Cohort					
1. Baseline Greedy CTC	15.43	79.52	28.92	50.20	0.40
2. CTC Prefix Beam Search	14.11	77.11	28.92	47.79	0.40
3. CTC + Blank Penalty ($\gamma = 0.2$)	15.25	79.12	28.92	49.40	0.80
4. CTC + Beam Search + LM	13.98	75.10	29.72	45.38	0.00
5. CTC + Beam + LM + Lexicon	13.80	74.70	29.72	44.98	0.00
6. Full Pipeline (Blank + Beam + LM + Lex)	13.58	73.09	29.72	43.37	0.00
Mild Dysarthria Cohort					
1. Baseline Greedy CTC	49.54	135.71	7.56	91.18	36.97
2. CTC Prefix Beam Search	46.45	140.76	7.56	91.18	42.02
3. CTC + Blank Penalty ($\gamma = 0.2$)	49.22	139.50	7.56	91.18	40.76
4. CTC + Beam Search + LM	47.69	120.17	17.65	80.67	21.85
5. CTC + Beam + LM + Lexicon	47.55	119.33	17.65	80.25	21.43
6. Full Pipeline (Blank + Beam + LM + Lex)	46.91	121.43	15.97	82.77	22.69
Moderate Dysarthria Cohort					
1. Baseline Greedy CTC	60.94	108.13	36.18	63.82	8.13
2. CTC Prefix Beam Search	57.07	107.72	36.18	63.82	7.72
3. CTC + Blank Penalty ($\gamma = 0.2$)	59.95	108.94	35.37	64.63	8.94
4. CTC + Beam Search + LM	57.70	100.41	44.31	55.69	0.41
5. CTC + Beam + LM + Lexicon	57.70	100.41	44.31	55.69	0.41
6. Full Pipeline (Blank + Beam + LM + Lex)	56.53	101.63	43.90	56.10	1.63
Severe Dysarthria Cohort					
1. Baseline Greedy CTC	90.28	100.00	76.28	23.72	0.00
2. CTC Prefix Beam Search	76.53	100.00	62.45	37.55	0.00
3. CTC + Blank Penalty ($\gamma = 0.2$)	88.26	100.00	76.28	23.72	0.00
6. Full Pipeline (Blank + Beam + LM + Lex)	77.52	100.00	80.63	19.37	0.00

3. **Severe Dysarthria (Blank-Token Deletion Regime)**: Extreme breathiness and vowel collapse produce low acoustic confidence, causing greedy CTC to default to blank tokens (76.28% deletion rate). Crucially, blank penalties and beam search address this through complementary mechanisms: suppressing the blank token alone via $\gamma = 0.60$ (Table 2) drives substantial CER reduction (90.28% $\rightarrow$ 79.54%) by raising non-blank emission probabilities, while prefix beam search (Table 3) preserves weak token hypotheses across paths, directly dropping deletions from 76.28% to **62.45%** and achieving 76.53% CER.

4.4 Empirical Validation on the TORGO Clinical Database

To validate whether our architectural findings generalize to real-world clinical pathology beyond parametric simulations, we evaluate our framework on the **TORGO database of dysarthric speech** [4]. TORGO consists of synchronized acoustic recordings from 15 participants across four clinical strata: 7 healthy adult controls, 2 mild dysarthria speakers (Cerebral Palsy and ALS), 2 mod-

erate dysarthria speakers (CP), and 4 severe dysarthria speakers (CP).

We structure a strict **speaker-disjoint cross-validation partition** partitioning all 15 participants:

- **Training Patient Cohort** (9 speakers): MC01, MC02, MC03, FC01, FC02 (Control); M03 (Mild, CP); F04 (Moderate, CP); M01, M04 (Severe, CP).
- **Held-Out Test Patient Cohort** (6 speakers): MC04, FC03 (Control); M05 (Mild, ALS); F03 (Moderate, CP); M02, F01 (Severe, CP).

Under this protocol, test speakers are completely unseen during training ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$).

Table 4 summarizes recognition performance across all four architectures on the held-out TORGO test patients.

The empirical results on real TORGO participants corroborate our theoretical and architectural hypotheses:

1. **Cross-Severity Interference Mitigation**: On control speech, decoupled routing and canonical anchor preservation cut CER from 61.81% down to **32.94%** (a -46.7% relative reduction), demonstrating that isolating severe dysarthric updates from healthy phonetic representations prevents negative acoustic interference.

Table 4: Clinical Speech Recognition Performance on Unseen TORGO Database Patients (Speaker-Disjoint Cross-Validation).

Model Architecture	Total Params	Control CER (%)	Mild CER (%)	Moderate CER (%)
Standard Single-Gate MoE	64,880	61.81	73.03	80.25
Matched Single-Gate (MLP Baseline)	74,790	45.77	68.54	77.78
Severity Router (Decoupled Only)	74,828	44.61	75.28	77.16
Calibrated Clinical MoE (Ours)	75,596	32.94	64.61	64.81

2. **Acoustic Calibration on Real Pathologies:** Speaker acoustic-average calibration provides substantial gains on unseen dysarthric speech, cutting Mild CER from 75.28% to **64.61%** (−10.67 pp) and Moderate CER from 77.16% to **64.81%** (−12.35 pp) on patients with Cerebral Palsy and ALS.

3. **Global Macro Improvement:** Overall, Calibrated Clinical MoE achieves the lowest Macro CER of **62.02%**, outperforming the standard single-gate MoE baseline by 14.11 absolute percentage points.

5 Conclusion

We have presented Neuromotor-Decoupled Clinical MoE with Acoustic-Average Speaker Calibration for atypical speech recognition. On unseen clinical patients evaluated under speaker-disjoint validation across both parametric cohorts and the 15-participant TORGO clinical database, decoupled severity routing reduces cross-severity interference, cutting typical CER by up to 53.1%, while zero-shot acoustic calibration reduces dysarthric CER by up to 12.35 percentage points on real patients. Furthermore, our progressive decoding ablation reveals that clinical speech recognition spans three distinct failure regimes across the severity spectrum, demonstrating that severity-conditioned routing must be paired with severity-specific decoding strategies. Future work will investigate upcycling pre-trained foundation audio models with calibrated neuromotor MoE layers.

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